



JPK

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PUSAT PENTADBIRAN KERAJAAN PERSEKUTUAN
62530 PUTRAJAYA**

KERTAS PENERANGAN
(*INFORMATION SHEET*)

KOD DAN NAMA PROGRAM / <i>PROGRAM'S CODE AND NAME</i>	Z-009-4: 2015 CORE ABILITIES	
TAHAP / LEVEL	4 (FOUR)	
NO. DAN TAJUK KEBOLEHAN TERAS / <i>CORE ABILITY NO. AND TITLE</i>	CA08 SYSTEM TECHNOLOGY APPLICATION	
NO. DAN PENYATAAN KEBOLEHAN / <i>ABILITY NO. AND STATEMENT</i>	08.01 ADOPT APPROPRIATE SOFTWARE APPLICATION 08.02 COMPLY TO SYSTEM ANALYSIS APPLICATION 08.03 UPDATE IT APPLICATION 08.04 ADHERE TO IT RULES & REGULATIONS	
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TAJUK/TITLE: IT APPLICATION UPDATE

TUJUAN/PURPOSE :

This Information Sheet is intended to describe on IT Application Update requirements to maintain and ensure information technology applications in the organisation are current.

The person who is competent in this CA shall be able to utilize information technology application in accordance with recommended practices.

PENERANGAN/INFORMATION:

1. What is System Software?

- 1.1 An operating system (OS) is a set of programs that manage computer hardware resources and provide common services for application software. The operating system is the most important type of system software in a computer system. Without an operating system, a user cannot run an application program on their computer (unless the application program is self booting).
- 1.2 Time-sharing operating systems schedule tasks for efficient use of the system and may also include accounting for cost allocation of processor time, mass storage, printing, and other resources.
- 1.3 For hardware functions such as input/output and main memory management, the operating system acts as a middleman between application programs and the computer hardware, although the application code is usually executed directly by the hardware it will frequently call the OS or be interrupted by it. Operating systems can be found on almost any device that contains a computer, from mobile phones and video game consoles to supercomputers and web servers.
- 1.4 Examples of popular modern operating systems include Android, iOS, Linux, Mac OS X and Microsoft Windows



Picture 1.4 Some of the existing OS on the market

1.5 You might hear the words Open Source when you read about operating systems. Linux is the most well known **Open Source OS** and Windows is the most well known **Closed Source OS**. So what does this mean?

- i) Open Source programs are programs where you can see the code (the source), you can edit it, copy it and use as you wish within the confines of an open source license. You can even sell it, but the person would be welcome to acquire the code for free if they wanted. As a result of this Linux is not developed by one person or one company but uses code developed by thousands of people, working for different organisations all over the world. And as a result there are many different versions (known as distributions) of Linux out there including Ubuntu, Slackware and Fedora. Android is built using Linux code.
- ii) Closed Source software doesn't allow people to look at the code or adapt it. Companies normally sell closed source products and you'll have to pay for a copy of Windows. Currently about 83% of desktops run the Windows OS.

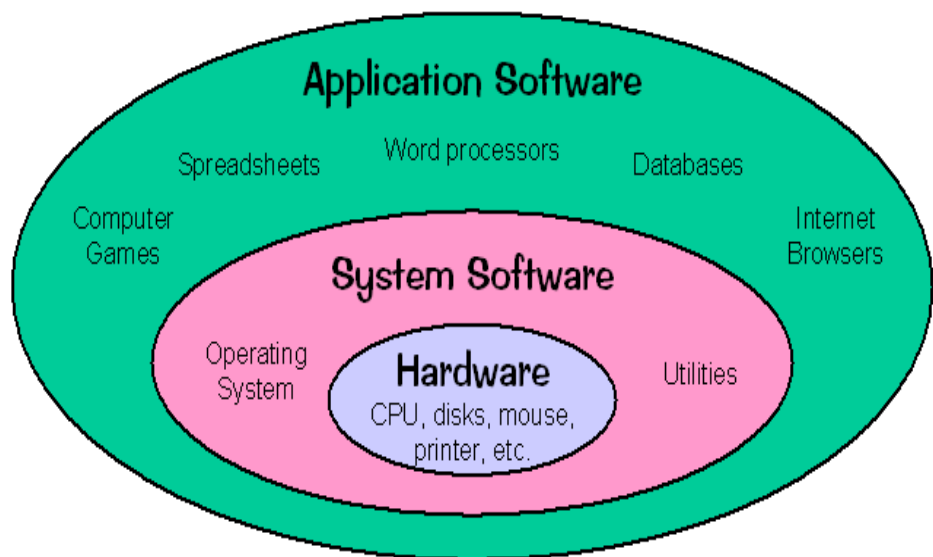
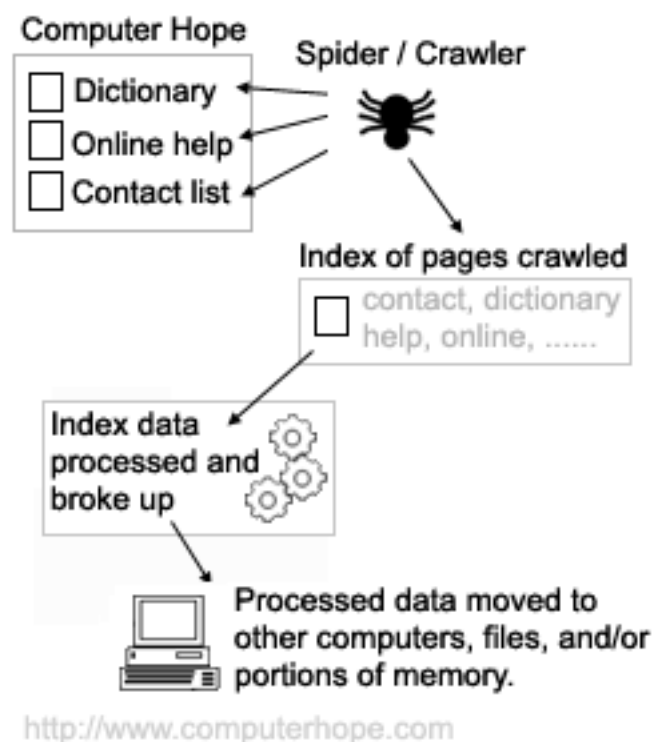


Diagram 3.5 Schematics of IT

2. What is search engines?

2.1 A search engine is a software program or script available through the Internet that searches documents and files for keywords and returns the results of any files containing those keywords. Today, there are thousands of different search engines available on the Internet, each with their own abilities and features. The first search engine ever developed is considered Archie, which was used to search for FTP files and the first text-based search engine is considered Veronica. Today, the most popular and well known search engine is Google.

2.2 How does it work? Because large search engines contain millions and sometimes billions of pages, many search engines not only just search the pages but also display the results depending upon their importance. This importance is commonly determined by using various algorithms.



Picture 6.2 How Search Engine works

2.3 As illustrated in the image on the right, the source of all search engine data is a spider or crawler, which automatically visits pages and indexes their contents. Once a page has been crawled, the data contained within the page is processed.

2.4 Often, this can involve the steps below.

- i) Strip out stop words.
- ii) Record the remaining words in the page and the frequency they occur.
- iii) Record links to other pages.
- iv) Record information about images or other embedded media.

2.5 The data collected above is used to rank the page and is the primary method a search engine uses to determine if a page should be shown and in what order.

2.6 Finally, once the data is processed it is broken up into one or more files, moved to different computers, or loaded into memory where it can be accessed when a search is performed.

2.7 Search Engines have been replacing dictionaries, encyclopedias and other knowledge references since the information is by the fingertips and returns the search results almost instantaneously.



Picture 2.7 Popular search engines

3. Internet Facilities

3.1 The Internet consists of tools that store, receive and transmit information electronically. Information and communication technology, which is a product of the Internet, has made it easier to access and share information than the conventional method of writing letters and sending them through the post office. Internet services such as instant messaging, Skype, email and teleconferencing allow users to communicate instantly with people from any part of the world at a relatively cheaper cost. These technologies allow people to collaborate on a project regardless of their geographical locations.



Picture 3.1 Communications through emails

3.2 The Internet has transformed the way of doing business in many fields. It has become a very vital tool for researchers seeking information and solutions to problems. The Internet has also made it possible for people to pursue education from any part of the world through online degrees and e-learning programs. Internet banking allows people to perform financial transactions without going to the bank.

3.3 The Internet has also increased job opportunities. For instance, organizations employ people to build and manage their websites and blogs through which they make their operations public to the world. Many institutional

libraries have taken advantage of the Internet to enhance users' access to information. For instance, the replacement of the manual catalogue with the electronic catalogue has increased efficiency in the search for books and journals in many libraries. The application of innovative Internet services in business such as health, education, commerce, banking and sports, has enhanced efficiency in service delivery. Social networking such as Facebook allows people to connect with friends and the rest of the world. Finally, people may spend their free time playing online games with other connected people.



Picture 4.3 Social media icons

3.4 Communication technologies provided using the Internet, such as emails, allow for an easy, cheap and simultaneous communication among many people from various parts of the world. Customized mailing groups and bulletin boards for groups of users have enhanced collaboration on various tasks such as homework. In addition, the Internet allows people to have access to a common folder or files, this enhances timely completion of projects.



Picture 3.4 Borderless world of IT

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SOALAN / QUESTION(S):

1. What is system software?

2. What is search engine?

3. What is internet facilities?

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